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Oxfordian magnetostratigraphy of Poland and its correlation to Sub-Mediterranean ammonite zones and marine magnetic anomalies

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ABSTRACT

A nearly continuous magnetostratigraphic polarity pattern was compiled from several ammonite-zoned carbonate successions of southern Poland and from a composite magnetostratigraphy from the Iberian Range of Spain. The array of sections spans the middle two-thirds of the Oxfordian within the Sub-Mediterranean Province (Cordatum through Bifurcatus ammonite zones). The average paleopole calculated from eight of these Polish sections is at 78.5°N, 184.9°E ($\delta p = 2.6°$, $\delta m = 3.5°$). The Sub-Mediterranean polarity pattern is consistent with an independent polarity pattern derived from the Boreal-realm sections of the British Isles, and improves the inter-correlation between these faunal realms. Cycle stratigraphy published for these ammonite subzones from southern France enabled temporal scaling of the polarity pattern, thereby facilitating correlation to marine magnetic anomalies M28 through M33 as modeled from deep-tow magnetometer surveys in the Western Pacific. The bases of the Middle and Upper Oxfordian substages as defined in the Sub-Mediterranean zonation in Poland correspond approximately to chrons M33 and M29 of that Pacific M-sequence model.

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1. Introduction

The birth of the present Pacific plate and the initiation of spreading in the North Atlantic and Indian Oceans began during the Middle to Late Jurassic (~175–145 Ma). The pattern of the M-sequence marine magnetic anomalies of latest Jurassic through early Cretaceous recorded at these new spreading centers (e.g., Larson and Hilde, 1975) has been successfully calibrated via magnetostratigraphic studies to biostratigraphy of the Kimmeridgian through basal Aptian stages (e.g., Channell et al., 1987; Ogg, 1988; Speranza et al., 2005). However, the models for the pre-M25 marine magnetic anomalies in the Pacific, which document the early growth of that plate (Handschumacher et al., 1988; Sager et al., 1998; Tivey et al., 2006; Tominaga et al., 2008), have remain uncalibrated owing to the lack of a complete and verified magnetic polarity pattern obtained from the outcrops for Bathonian through Oxfordian.

Kądziałko-Hofmokl and Kruczyk (1987) investigated the magnetostratigraphy of several Middle–Upper Jurassic outcrops in southern Poland and suggested that the Middle Callovian sediments might correlate with the Pacific anomalies M27 or M28. During the 1990s, additional magnetostratigraphic studies of outcrops in Poland (e.g., Ogg et al., 1991; Kruczyk et al., 1995) and additional sections in Spain yielded a skeleton pattern of magnetic polarity zones, and Pacific anomalies M27–M28 were correlated instead to the Middle Oxfordian (Ogg and Gutowski, 1996). However, the stratigraphic coverage was limited because of the poorly determined ammonite zones, the weak magnetization of some intervals were at the noise level of that generation of magnetometers, and the results were not checked by the rock-magnetic studies. Subsequently, the compilation and verification of a detailed reference scale for the Middle to Late Jurassic geomagnetic history has lain dormant.

We derived a nearly continuous magnetic polarity pattern of the Lower through lower-Upper Oxfordian (Cordatum–Bifurcatus ammonite zones) from ten Sub-Mediterranean sections of cratonic southern Poland and from a composite of the NE Iberian Range in Spain (Section 2.3). These magnetostratigraphies are tied to precise biostratigraphic horizons within the high-resolution Sub-Mediterranean Oxfordian zonation of Poland (e.g. Table B1.1 in Matyja and Wierzbowski, 2006a). A coeval stratigraphic interval in southern France has yielded a cycle stratigraphy for calculating the duration of most subzones of the French Oxfordian scheme (Boulila, 2008; Boulila et al., 2008, in press). Those results were applied in this study to estimate the durations of magnetic polarity zones, thereby enabling us to correlate the magnetostratigraphic polarity pattern to models of marine magnetic anomalies M28 through M33.

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2. Stratigraphy of sampling sites

2.1. Polish sections

In the Late Jurassic, epicratonic marine basins extended along the northern margin of the Tethys Ocean (e.g. Ziegler, 1990). The Polish Jura Chain and the Holy-Cross Mountains belonged to the southern margin of the Central European Basin in the Sub-Mediterranean Province of the peri-Tethyan biogeographic realm (Kutek, 2001; Matyja and Wierzbowski, 2006b). During the Oxfordian, extensive sponge-rich carbonate facies attained a spectacular thickness of some hundred meters (Kutek, 1968; Matyja, 1977; Kutek et al. 1992; Matyja and Wierzbowski, 1996). These Oxfordian limestones are exceptionally useful for paleontological research, due to their continuous sedimentation, homogeneous facies, and richness in ammonites (Brochwicz-Lewiński, 1973, 1976; Głowniak and Wierzbowski, 2007; Kutek, 1968; Malinowska, 1972; Wierzbowski, 1978). During the Oxfordian, the southern Poland basin received variable incursions of Boreal and Mediterranean ammonites, thereby making this area important for the inter-regional correlations (Atrops et al., 1993; Głowniak, 2006c; Kutek et al., 1984; Matyja and Wierzbowski, 1996; Matyja et al., 2006).

The Polish Jura Chain (Fig. 1C) is a part of the Fore-Sudetic Monocline (Znosko, 1998). The Oxfordian carbonate succession consists of (1) the Lower Oxfordian *Jasna Góra Beds* of alternating marly limestones and marls, rich in benthic and pelagic fauna (mainly sponges and ammonites), and (2) the middle Oxfordian *Zawodzie Beds* of thick-bedded limestones, laterally interfingering with cyanobacterial-sponge bioherms (Różycki, 1953; Matyja and Wierzbowski, 1996, 2006b). The overlapping sections of Ogrodzieniec (OG), Wysoka (WY), Niegowonice (NE) and ZawodzieII (ZD) in the vicinity of Częstochowa span the Cordatum Zone of Lower Oxfordian through the Bifurcatus Zone of Upper Oxfordian, but with a small sampling gap at the Buckmani/Elisabethae subzone boundary within the Transversarium Zone. The sections of Podłęże1-3 (PD), Rudno (RD) and Zalas2 (ZA), about 60 km west of Kraków, span portions of the Lower to Middle Oxfordian.

The Holy-Cross Mountains is a part of the Mid-Polish Anticlinorium structure that formed in latest Cretaceous or Paleocene (Znosko, 1998; Kutek, 2001; Gutowski, 2006; Gutowski and Koyi, 2007). The Oxfordian limestones have three main units (after Matyja, 1977): (1) basal *Jasna Góra Beds*, with an average thickness of about 5 m; (2) micritic to wackestone *Gray Limestones* devoid of shelly macrofauna, with dispersed iron sulfide responsible for its gray color; average thickness of ca. 15 m; (3) thick-bedded *Morawica Limestones* with ammonites and siliceous concretions; thickness up to 100–150 m. An Oxfordian–Callovian boundary discontinuity encompasses the basal Oxfordian Mariae Zone (Drewniak and Matyja, 1992). The two lower Oxfordian sections of Gnieździska quarry (GN and GN2) are on the opposite limbs of the Mnin fold. The section of Wola Morawicka (WM), ca. 20 km south of Kielce, overlaps the upper part of GN and extends into the uppermost Transversarium and/or Bifurcatus Zones.

2.2. Sub-Mediterranean Oxfordian zonation in Poland

The Sub-Mediterranean Oxfordian zonal scheme used in this paper has only recently been comprehensively described (e.g., by Matyja and Wierzbowski, 2006a, their Table B1.1). Our magnetostratigraphy sections of Niegowonice, Ogrodzieniec, Wysoka and ZawodzieII are the primary reference sections for the Middle–lower Upper Oxfordian

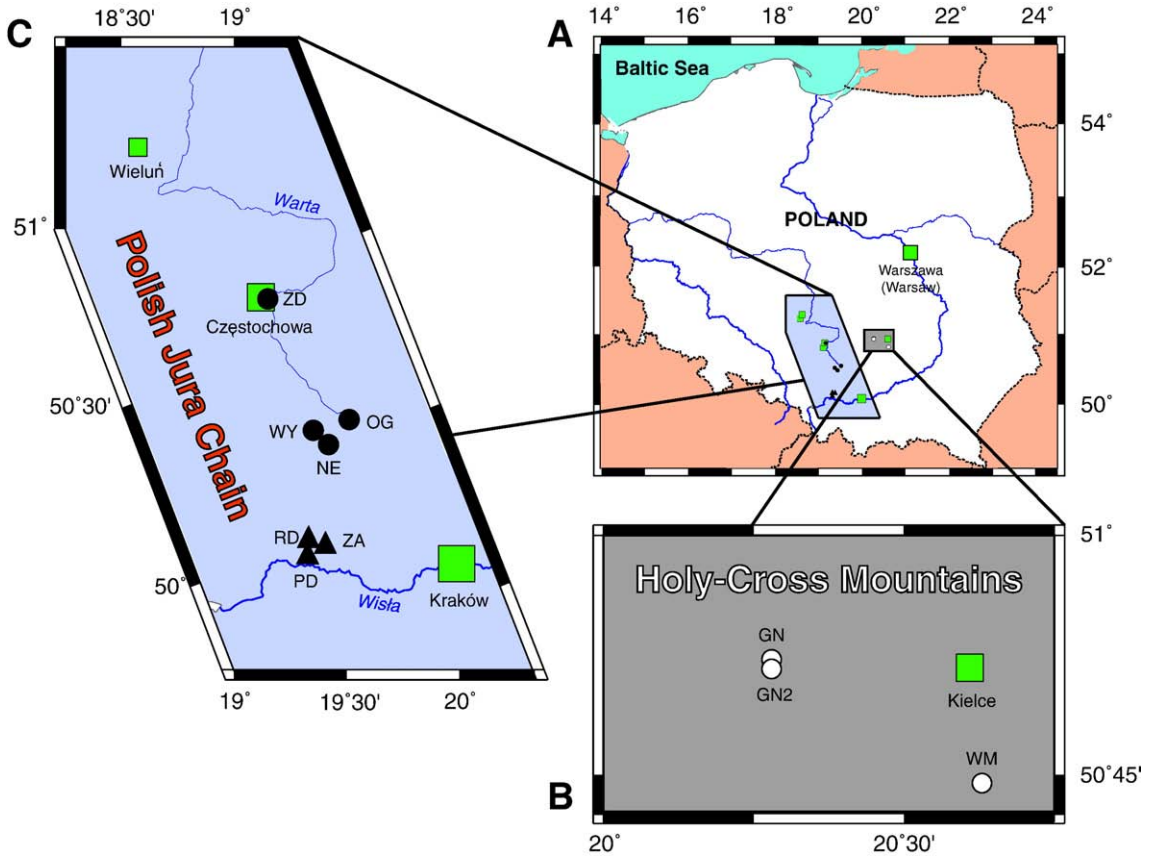


Fig. 1. Locations of the outcrops from the cratonic region of southern Poland sampled for magnetostratigraphy for this study are indicated by black and open circles: Niegowonice (NE), Ogrodzieniec (OG), Wysoka (WY), and ZawodzieII (ZD) in the Polish Jura Chain (Fig. 1C); and Gnieździska (Góra Dybkowa GN and Góra Maćkowa GN2) and Wola Morawicka (WM) in the SW margin of the Holy-Cross Mountains (Fig. 1B). We also used the primary data to re-interpret previous results from sections that had been published by Ogg et al. (1991) and Ogg and Gutowski (1996), shown here as black triangles: Podłęże-3 (PD), Rudno (RD), Zalas2 (ZA). See Table 1 for further details on individual sections.

of this zonation (Ogrodzieniec: Dembicz et al., 2006; Matyja and Głowniak, 2003; Wysoka: Głowniak, 2002; Niegowonice: Głowniak, 2006a; ZawodzieII: Matyja and Wierzbowski, 1996; Głowniak, 2006b: Table 1).

The lowermost Oxfordian Scarburgense and Praecordatum subzones of the Mariae Zone are condensed throughout Poland, but the Cordatum Zone (Bukowskii, Costicardia and Cordatum subzones) of Lower Oxfordian is well established. The subdivisions of the Plicatilis Zone (Paturattensis, Ouatius and Arkelli subzones) and Transversarium Zone (Buckmani and Elisabethae subzones) of Middle Oxfordian accommodate two successive natural assemblages of a single lineage of the genus *Perisphinctes* (Głowniak and Wierzbowski, 2007; Głowniak et al., 2008). However, the Plicatilis–Transversarium boundary as defined in Poland at the level of phyletic first appearance of the earliest species of the subgenera *Dichotomosphinctes* and its macroconch counterpart of *Perisphinctes* s.str. is slightly older than the definition of the Plicatilis–Transversarium boundary in Western and NW Europe (Głowniak, 2006c versus Callomon, 1960).

The Middle–Upper Oxfordian boundary placed in Poland at the base of the Wartae Subzone of the Bifurcatus Zone is also slightly older than the definition of the Bifurcatus Zone in France (Głowniak, 2006a versus Cariou et al., 1997). The Upper Oxfordian, as used in Poland, accommodates the youngest natural assemblages of the *Perisphinctes* lineage: the species of the subgenus *Dichotomoceras* and its macroconch counterparts of *Perisphinctes* s.str. The lower boundaries of the successive Stenocycloides and Grossouvrei subzones in the Bifurcatus Zone are coeval in Poland and in Western Europe. The higher Bimammatum and Planula zones of the Upper Oxfordian (partly equivalent to the Lower Kimmeridgian sensu Boreal) and their correlations in Europe are described in Przybylski et al. (in press).

2.3. Magnetostratigraphic reference sections of Spain

Several paleomagnetic studies within the NE Iberian Range in Spain compiled and verified a magnetostratigraphic composite pattern (Steiner et al., 1986; Juárez et al., 1994, 1995) for portions of the Middle and Upper Oxfordian. The composite spans the Luciaeformis Subzone of the Spanish version of the Transversarium Zone into the Hypselum Subzone of the Bimammatum Zone (biostratigraphy by Meléndez, 1989; Fontana and Meléndez, 1990; Meléndez and Fontana, 1991; Pérez-Urresti, 1994). Our biostratigraphic calibration of this polarity pattern differs in certain intervals from the version published in Juárez et al. (1995) by incorporating the possible stratigraphic gaps reported at the boundaries of the Schilli–Rotoides and Rotoides–Stenocycloides subzones (upper Transversarium and lower Bifurcatus zones, respectively, as used in Spain). These hiatuses were described by Bello et al. (1995) and mentioned by Meléndez and Fontana (1991). An independent argument for these possible stratigraphic gaps derives from the comparison with complete ammonite succession from Poland (Głowniak, 2006a).

3. Analysis of paleomagnetic samples

3.1. Demagnetization procedure and magnetic behavior

Paleomagnetic samples from the different field campaigns in Poland were analyzed in four laboratories (magnetically shielded rooms at the University of Wyoming and the University of Michigan in USA, and Helmholtz-coil field-free spaces at the GeoForschungsZentrum Potsdam in Germany and the Polish Academy of Sciences). Remanent magnetization was measured using two- or three-axis cryogenic magnetometers. Programs of progressive demagnetization by thermal steps

Table 1
Sub-Mediterranean sections of the uppermost Callovian–early Upper Oxfordian sections — tectonic location (HCM — Holy Cross Mountains, PJCh — Polish Jura Chain), position in latitude–longitude (see Fig. 1), biostratigraphic interval, average bedding orientation, number of samples, year of sampling, paleontology references, and previous magnetostratigraphy studies. The main field campaigns for magnetostratigraphic sampling of the Polish sections were pilot studies in 1982, 1992–93 by J.G. Ogg, followed by an extensive 2006–07 suite by P.A. Przybylski. Samples were drilled generally every 0.5–1 m in the Middle and Upper Oxfordian and denser in the condensed sections of the Lower Oxfordian. Abbreviations of section names are explained in the caption of Fig. 1.

Section	Location	Position		Biostratigraphic span of sampling	Average bedding		Analyzed samples	Year of sampling	References	
		Lat [°]	Lon [°]		Dip [°]	Dip Dir [°]			Biostrat	Magnetostrat
WM*	HCM	50.74	20.63	Transversarium? – Bifurcatus?	16	235	19	1993	Matyja, 1977	Ogg and Gutowski, 1996
ZD	PJCh	50°48′N	19°09′E	Bifurcatus	2	92	29	1992, 1993	Brochwicz-Lewiński 1973, 1976; Głowniak, 2006a,b; Malinowska 1972	Ogg and Gutowski, 1996
NE	PJCh	50°23′N	19°25′E	Transversarium – Bifurcatus	39	317	12	1992	Głowniak, 2006a; Matyja and Wierzbowski, 1996	Ogg and Gutowski, 1996
WY	PJCh	50°26′N	19°21′E	Cordatum–Transversarium	6	166	48	1992, 2006, 2007	Głowniak, 2002	Ogg and Gutowski, 1996
OG	PJCh	50°28′N	19°31′E	Cordatum–Transversarium	3	219	33	2006, 2007	Dembicz et al., 2006; Głowniak, 2002; Matyja and Głowniak, 2003	Kądziałko-Hofmokl and Kruczyk, 1987
GN2**	HCM	50°52′N	20°17′E	Cordatum–Bifurcatus?	50	35	55	1993, 2006, 2007	Głowniak 2008, 2009	Kruczyk et al., 1995
GN	HCM	50°53′N	20°17′E	Cordatum–Plicatilis	16	229	77	1992, 1993, 2006	Drewniak and Matyja, 1992; Głowniak, 2008, 2009	Kruczyk et al., 1995
PD (1–3)	PJCh	50°02′N	19°33′E	Plicatilis–Bifurcatus	8	251	52	1982	Żywiecki 1993	Ogg et al., 1991; Ogg and Gutowski, 1996
RD	PJCh	50°06′N	19°34′E	Plicatilis	5	40	10	1982	Głowniak and Wierzbowski, 2007; Hoffman in Ogg et al., 1991;	Ogg et al., 1991; Ogg and Gutowski, 1996
ZA	PJCh	50°05′N	19°37′E	Cordatum–Plicatilis	7	30	26	1982	Matyja 2006; Matyja and Tarkowski, 1981	Ogg et al., 1991; Ogg and Gutowski, 1996
Composite of 6 sections	NE Iberian Range, Spain	41°04′N	1°00′W	Transversarium–Bimammatum	49	147	408	1993, 1994	Bello 1994; Bello et al., 1995; Meléndez, 1989; Meléndez and Fontana, 1991	Juárez et al., 1994; Juárez et al., 1995; Steiner et al., 1986

273 of total 361samples from Poland used for the present study.
Over 100 samples were selected to compute the upper-Lower–lower Upper Oxfordian paleopole.
*Section not used in the analysis due to the not available biostratigraphy at present.
**Section not used in the analysis due to a secondary overprint.

and/or by applying an external alternating field (AF) were decided for each lithology after pilot studies. Most samples were demagnetized at 25–30 °C thermal steps and/or 5–10 mT AF steps within the typical range of characteristic directions. Bulk susceptibility was generally monitored after every second thermal demagnetization step below 300 °C and after every temperature step above 300 °C.

Even though magnetic characteristics varied greatly among the various lithologies at each outcrop, there were some general

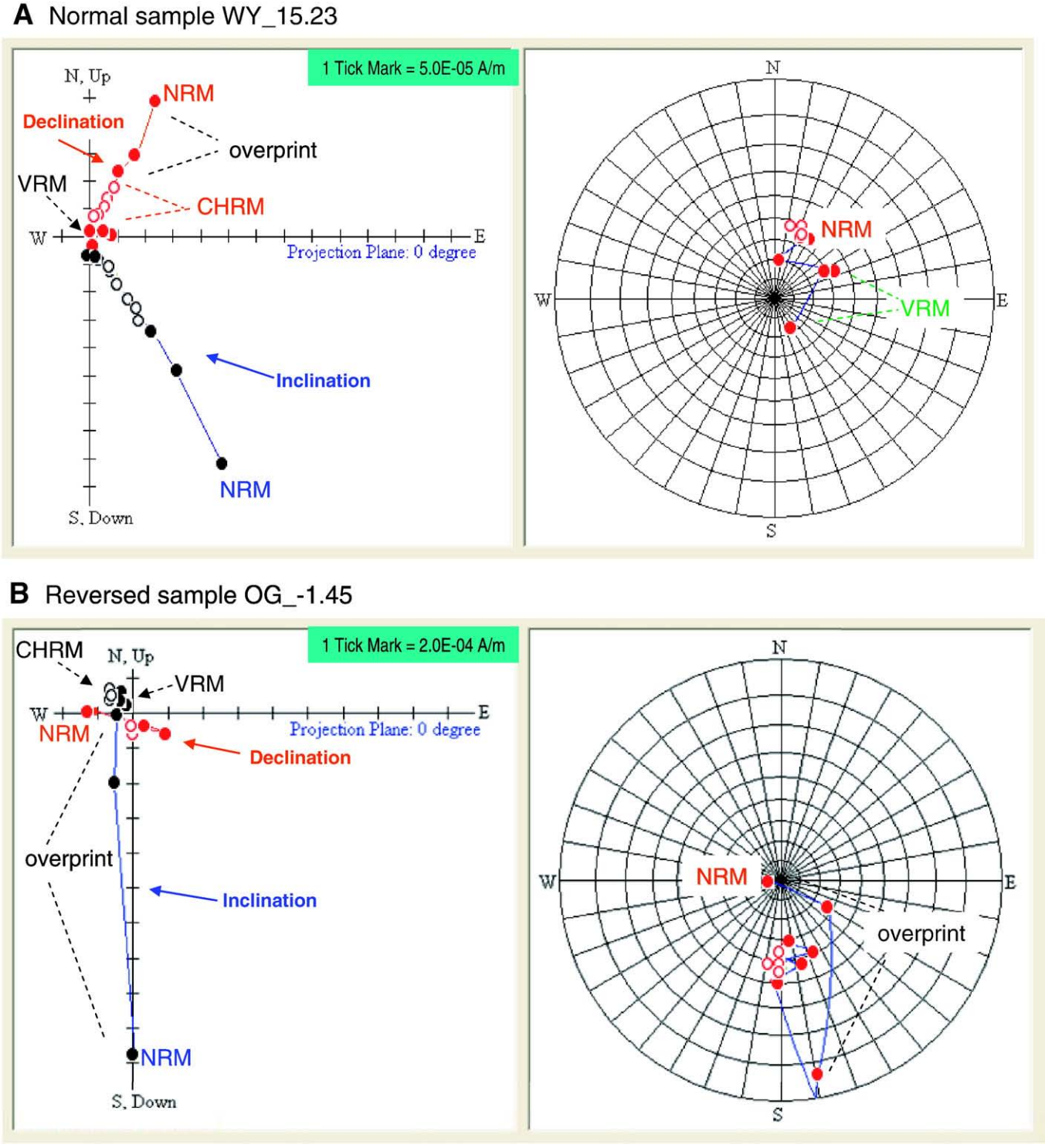


Fig. 2. Typical thermal demagnetization behavior of samples interpreted as normal- and reversed polarity on vector diagrams (left side of each pair) and stereoplots (right). (A) Normal-polarity sample WY_15.23 is from the spongy limestone bioherm at Wysoka; characteristic vector (6 thermal steps, 275°–400° = 25.1° declination, 55.5° inclination, 1.2×10^{-4} A/m intensity) is rated "N" based upon its tight clustering and uni-vectoral decay. (B) Reversed-polarity sample OG_-1.45 is from a platy limestone at the Ogrodzieniec quarry; characteristic vector (4 thermal steps, 325°–400° = 180.8° declination, −48.2° inclination, 1.6×10–4 A/m intensity) is rated "R" based upon its tight clustering and uni-vectoral decay. Declination and inclination vectors correspond to the horizontal component and to the actual dip of the magnetization vector; vectors have been corrected for tilt of strata; units of intensity are in A/m. Thermal steps selected to compute the characteristic directions are marked as open circles. Interpreted directions of secondary magnetizations and VRM ranges are shown by dashed lines or arrows. All measurements are in the supplemental data appendix.

characteristics. Initial NRM directions of the limestones from Poland are typically dominated by normal-polarity oriented parallel with the present dipole field. This secondary overprint was largely removed upon heating above 200°–300 °C, then divergent polarities among samples were typically observed through 400 °C. Above 400 °C, the remanent magnetic intensities of samples had usually decreased to near the noise level of the magnetometers or a surge in susceptibility or spurious data indicated possible onset of viscous remanent magnetization (VRM). Despite performing the thermal demagnetization and measurements within a shielded room, the onset of spurious viscous magnetizations generally rendered measurements beyond 400 °C to be random noise. Demagnetization sets for each sample with annotations, characteristic directions and interpretations are in the Supplementary data appendix to this paper.

The typical magnetic behavior during thermal demagnetization suggests that a common carrier of characteristic magnetizations (generally

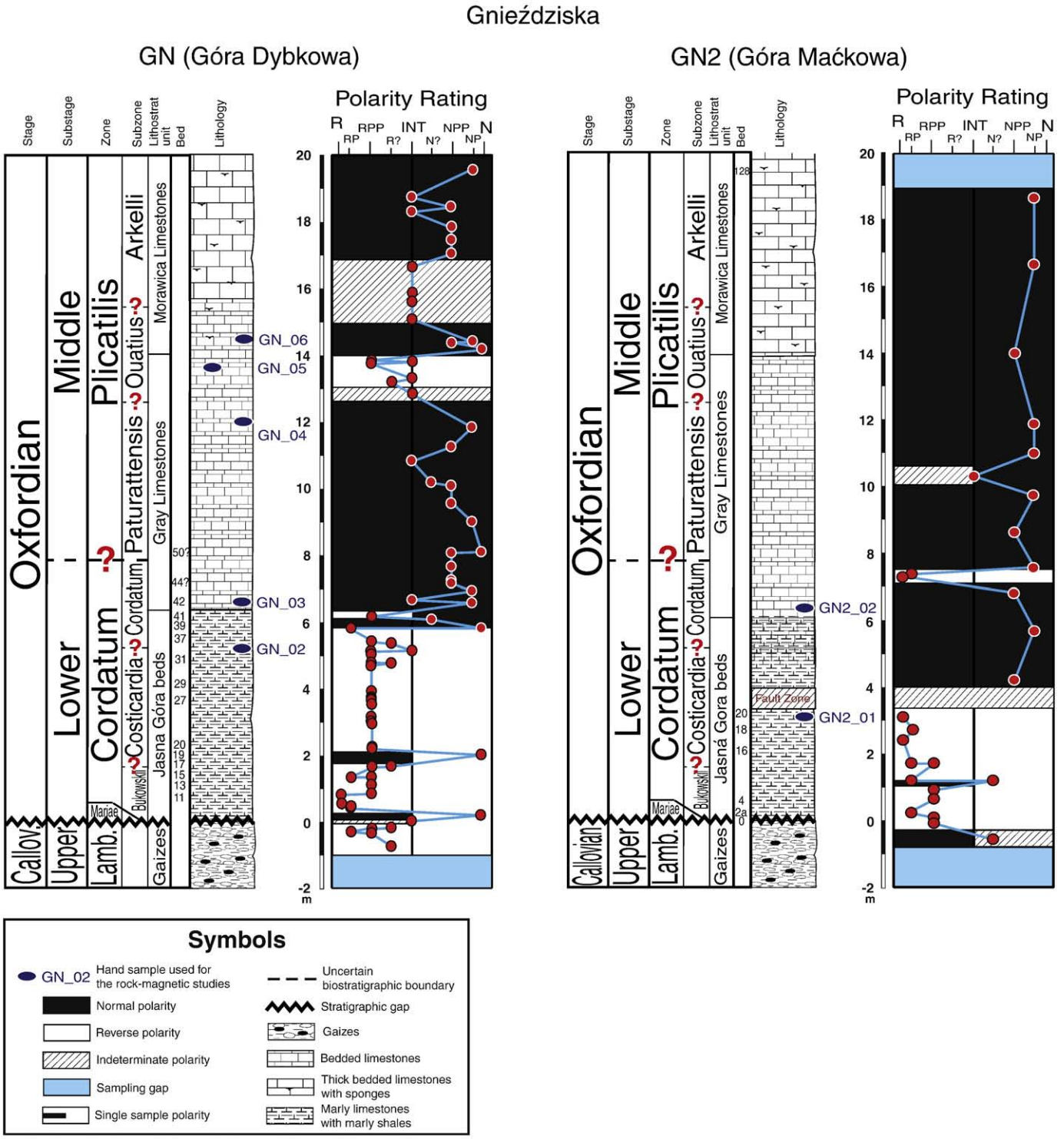


Fig. 3. Rock-magnetic sampling (black ovals on lithology column) and interpreted magnetostratigraphy in the coeval strata of two sections of Gnieździska quarry: GN (hilltop of Góra Dybkowa) and GN2 (hilltop of Góra Maćkowa), which are on the opposite limbs of the Mnin fold. The active quarry section of GN2 continues upward by over 100 m and terminates within the Upper Oxfordian Bifurcatus Zone. Unfortunately, this faulted GN2 section revealed a diagenetic overprint, therefore its magnetostratigraphy is not considered reliable (see text). The precise location of the zonal and subzonal boundaries in the higher parts of these sections is currently under study.

the 200 °C–400 °C range) is magnetite. The surge in susceptibility above 350°–400 °C observed in many samples is probably due to a combination of dehydration of iron-rich clays and the oxidation of iron sulfides.

3.2. Polarity interpretation

Analysis of characteristic directions was mainly performed using the *Paleomag* software (Zhang and Ogg, 2003). About 20 samples were analyzed independently by Marek Lewandowski using his *Paleomagnetic Data Analysis* software (Lewandowski et al., 1997). Calculation of characteristic directions used a three-dimensional least-squares line-fit method for user-selected or software-selected steps (Kirschvink, 1980). Characteristic directions were generally assigned within the 200°–400 °C interval, in which most samples either exhibited stable paleomagnetic directions through multiple heating steps as the intensity of magnetization decreased (Fig. 2), or developed a semi-stable direction prior to onset of VRM at higher temperatures. The magnetic behavior, the selection of steps for the characteristic direction and the polarity rating for each sample were analyzed independently by two observers (three independent observers for part of section GN2), and a compromise conservative assignment was chosen after discussing the interpretations.

Each characteristic direction was assigned a polarity rating based upon the individual demagnetization behavior: (1) N or R implies well-defined directions computed from at least three successive demagnetization steps within 15° of each other and accompanied by a simultaneous decrease of magnetic intensity, (2) less-precise NP or RP have directions computed from a set of steps that differed by more than 15°, or if the characteristic direction was determined from only two steps, (3) NPP or RPP-rated samples had obvious polarity trends, but with either more dispersed direction or those that did not achieve adequate cleaning during demagnetization; these were omitted from mean pole computations, (4) and samples with uncertain (N? or R?) or indeterminate (INT) polarity that were not used to define polarity zones.

4. Rock-magnetic study of the origin of the paleomagnetic reversals of Gnieździska sections

4.1. Methods

Most of our sections yielded similar polarity patterns within the same ammonite subzone; and this reproducibility gave us confidence that the selected characteristic directions were consistent with a primary Jurassic signal. However, the apparent polarity patterns from two nearby sections of the Gnieździska quarry, Góra Dybkowa (GN) and Góra Maćkowa (GN2) were inconsistent for the presumed coeval strata (Fig. 3). Therefore, we undertook a comparative analysis of the rock-magnetic properties of these two sections in intervals with apparent high-rated normal and reversed polarity (N/R) and at levels above and below some of the inferred polarity transitions. It is known that some reversals may be erroneously assigned of being primary in age, while in fact they stem from a secondary overprint due to diagenetic changes affecting the rock (e.g. Machel, 1995). The geochemical environment may be altered by fluid migration, elevated temperature due to burial, or other processes. As a consequence, new magnetic phases can be created due to oxidation or reduction processes that record an ambient geomagnetic field that post-dates the host rock deposition. Bearing this in mind, we performed rock-magnetic tests based on a simple assumption that, for a primary origin of the paleomagnetic record, there should be no correlation between observed polarity pattern and assembly of magnetic carriers.

We identified assemblages of magnetic carriers by utilizing the Lowrie test (Lowrie, 1990) and scanning electron microscopy (SEM) with EDX and WDX (energy- and wavelength-dispersive X-ray) electron microprobe. This rock-magnetic study was performed in the laboratories of the Institute of Geophysics at the Polish Academy of Sciences and in the Department of Geology at the University of Warsaw (both in Warsaw, Poland).

The common ferromagnetic minerals have distinctive, characteristic coercivities and thermomagnetic properties. The analysis of the acquisition curve of isothermal remanent magnetization (IRM) combined with subsequent thermal demagnetization of the IRM gives a clear interpretation of the ferromagnetic mineral content of a rock (Lowrie, 1990; Suppl. Fig. 1). Subsequently, different coercivity fractions of IRM were remagnetized in successively smaller fields along three orthogonal directions *Z*, *Y* and *X* with the corresponding magnetic fields 2.8 T, 0.4 T and 0.12 T. Each orthogonal component of the composite IRM was monitored during progressive thermal demagnetization (Fig. 4). Analysis of the hysteresis parameters (the Day plot, Day et al., 1977; Suppl. Fig. 2) was useful in the evaluation of the grain sizes of the magnetic carriers.

The rock samples (ca. 2 kg) were dissolved in an acetic acid (Hounslow and Maher, 1999), and magnetic minerals were extracted from the insoluble fraction by means of a neodymium magnet placed in a removable, thin latex-made envelope. The separated magnetic grain fractions and their element spectra were investigated using a scanning electron microscope (Suppl. Fig. 3).

4.2. Interpretation

Magnetic grain sizes are within the limit of SD/PSD states, being therefore capable of carrying the ancient magnetization (Suppl. Fig. 2). The Lowrie test for section GN indicated a presence of both magnetite and hematite in variable ratios, but didn't exclude the possibility that some intervals interpreted as a normal-polarity magnetization may have been an artifact of secondary hematite that formed during a later oxidation event (Fig. 4).

However, the Lowrie test for section GN2 reveals a more significant difference between magnetic characteristics of the samples of opposing polarity (Suppl. Fig. 1, Fig. 3, Fig. 4), suggesting that they may not record a geomagnetic field of the same age. Consequently, we are uncertain which of the two magnetic polarities (if either) is of primary origin. In particular, fluids migrating along a fault plane could be responsible for a chemical remagnetization of the original Late Jurassic component at least in one of the walls. This hypothesis would be consistent with the SEM analysis of the insoluble fractions (summary given in caption to Suppl. Fig. 3). These rock-magnetic results, coupled with the anomalous apparent polarity pattern when compared to coeval sections that had greater sampling densities, led us to exclude the entire GN2 section from the magnetostratigraphic composite.

Iron sulfides are much less abundant in section GN, and the presence of detrital Ti magnetite and magnetite grains supports the hypothesis of a primary origin of its characteristic magnetization (Supp. Fig. 3). However, the suspicious results from one sample led us to exclude a small interval from the magnetostratigraphy composite.

Fig. 4. Thermal demagnetization of a three-component IRM (Lowrie test) imposed on the samples from the two Gnieździska sections (GN and GN2). (1) section GN (upper 4 charts). The Lowrie test indicated a presence of both magnetite (T_{ub} ca. 560 °C) and hematite (T_{ub} ca. 670 °C) in variable ratios. For example, samples GN_01 and 02 from the sample GN_02 (reversed polarity) have magnetite as the dominant magnetic carrier and display the same unblocking temperature for the hematite component, but the less rectangular shape of the IRM acquisition curve for the GN_02 sample is indicative of a relatively larger hematite content. In contrast, overlying sample GN_03 and GN_04 from the same normal-polarity interval have a dominant hematite carrier (Suppl. Fig. 1, Fig. 3). Samples GN_05 (Reversed) and GN_06 (Normal) are similar in magnetic minerals composition to the sample GN_02 (Reversed). Therefore, it is possible that the characteristic magnetization of normal-polarity for GN_03 represents the magnetization of secondary hematite that formed during a later event of secondary oxidation. (2) section GN2 (lower 2 diagram). Samples interpreted as reversed polarity" (GN2_01) and normal polarity (GN2_02) are from different walls of a local fault zone.

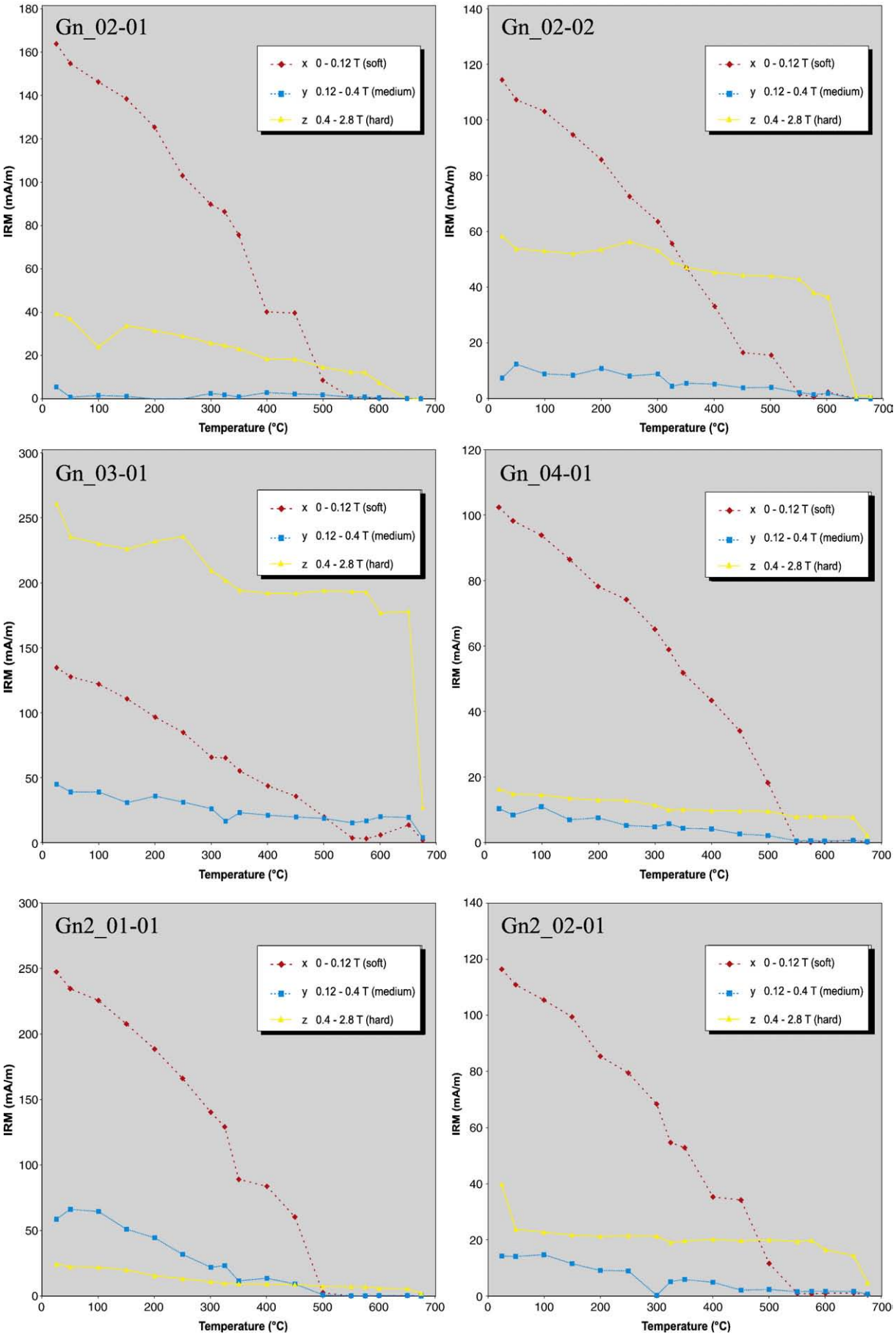

Gn_02-01
x 0 - 0.12 T (soft)
y 0.12 - 0.4 T (medium)
z 0.4 - 2.8 T (hard)
IRM (mA/m)
Temperature (°C)
Gn_02-02
x 0 - 0.12 T (soft)
y 0.12 - 0.4 T (medium)
z 0.4 - 2.8 T (hard)
IRM (mA/m)
Temperature (°C)
Gn_03-01
x 0 - 0.12 T (soft)
y 0.12 - 0.4 T (medium)
z 0.4 - 2.8 T (hard)
IRM (mA/m)
Temperature (°C)
Gn_04-01
x 0 - 0.12 T (soft)
y 0.12 - 0.4 T (medium)
z 0.4 - 2.8 T (hard)
IRM (mA/m)
Temperature (°C)
Gn2_01-01
x 0 - 0.12 T (soft)
y 0.12 - 0.4 T (medium)
z 0.4 - 2.8 T (hard)
IRM (mA/m)
Temperature (°C)
Gn2_02-01
x 0 - 0.12 T (soft)
y 0.12 - 0.4 T (medium)
z 0.4 - 2.8 T (hard)
IRM (mA/m)
Temperature (°C)

Table 2
(A) Oxfordian mean directions and paleopole from Poland. The reversal tests of Butler (1998) and of McFadden and McElhinny (1990) test if the mean normal and reversed directions of the samples belong to the same population, i.e. if the magnetic signals recorded from each sample represent the same time in the geological history. However, the McFadden and McElhinny reversal test requires more than five highest-quality samples of each polarity, but there were generally inadequate samples for each polarity at most outcrops. Therefore, all data were merged from the eight sections for these statistical tests (Suppl. Fig. 4). The dual polarities of this composite set passed both significance tests and obtained class A of the McFadden and McElhinny test. (B) Comparison with other published Late Jurassic poles from Europe.

A. Mean directions and paleopole of the Lower–Middle Oxfordian and significance tests of data set

Biostratigraphic span of sampling	Section	Location	Magnetic directions	After bedding correction							Mean virtual geomagnetic pole							Reversal tests	
				Least squares (weighted)														Butler (1998)	McFadden and McElhinny (1990)
				Number	Decl (°)	Incl (°)	Intensity (A/m)	α95 (°)	K	R/N	Lat (°)	Long (°)	δp (°)	δm (°)	Paleo Lat (°)	upper (+)	lower (−)	F/Ft	γ/γc
Lower–Middle Oxfordian (Cordatum–Bifurcatus)	Composite of 8 sections (ZD, NE, PD, WY, OG, RD, ZA, and GN)	Poland, Polish Jura Chain and Holy Cross Moutains	Normal	52.5	1.8	59.8	1.40E−04	3.1	41.0	51.24/52.5	80.1	191.8	3.5	4.7	40.7			1.52/3.04	3.37/4.67
			Reverse	47.0	185.9	−57.2	5.62E−05	3.6	35.0	45.69/47.0	−76.6	−0.7	3.8	5.2	−37.8				
			Combined	99.5	3.8	58.6	9.09E−05	2.3	37.7	96.89/99.5	78.5	184.9	2.6	3.5	39.3	41.9	36.8	Positive test	Positive test, class A

B. Comparison of mean directions and pole position with previous studies

Location	Stratigraphic age	Sites	Decl (°)	Incl (°)	α95 or A95 (°)	K	Lat (°)	Long (°)	Paleo Lat (°)	upper (+)	lower (-)	References
Southern Poland	Lower–Upper Oxfordian	11	3.8	58.5	2.3	37.7	78.5	184.9	39.2	41.8	36.7	Present study
Polish Jura Chain	Oxf.–Kim. bound.	5	6.5	58.0	2.8	58.4	77.4	175.9	38.6	41.7	35.7	Przybylski et al. (in press)
European Craton	Oxf.–Kim. bound.	14	5.8	55.4	3.3	782.1	75.0	181.2	36.0	38.8	33.3	
Europe from APWP	Uper Kimmeridgian (mean age)	10	12.1	57.5	6.6	54.3	75.0	159.9	38.1	45.8	31.6	Besse and Courtillot (2002)
Europe from APWP	Upper Callovian (mean age)	15	18.6	58.1	5.0	59.7	72.5	144.0	38.8	44.6	33.7	

Abbreviations:

Number, *N*	total number of observations (specimens) used in the analysis (fractional if data is weighted)
Decl	mean geographic declination
Incl	mean geographic inclination
Intensity	mean intensity of the ChRM directions
α95 [°]	angular radius of the cone of 95% confidence level
K	Fisher precision parameter
R	length of the resultant vector of all individual directions
Lat, Lon	latitude and longitude of the virtual geomagnetic paleopole
δm	radius of the 95% confidence level ellipse of the paleomagnetic pole, measured along the direction of the magnetic meridian from the site to the vertical pole
δp	radius of the 95% confidence level ellipse of the paleomagnetic pole, measured perpendicular to the direction of the magnetic meridian from the site to the vertical pole
PaleoLat magnetic	paleolatitude of the sampling location
+/− on PaleoLat	upper and lower confidence limits on the magnetic paleolatitude; mean values of these limits are given
F	Fisher statistic for reversal test (Butler 1998)
Ft	tabulated *F* value for $p=0.05$ for 2 and 2(*N*-2) degrees of freedom
γ	spherical angle between the mean normal and mean inverted reverse direction
γc	reversal critical angle (McFadden and McElhinny, 1990)
Sites	number of localities.

5. Paleomagnetic directions and paleopoles

5.1. Calculation of mean paleomagnetic directions and antipodal tests

Samples classified as N, R, NP and RP (Suppl. Fig. 4) were used to calculate mean directions, paleomagnetic poles and paleolatitudes. Mean directions were calculated separately for normal-, reversed- and combined-polarity samples using a Fisher method (Fisher, 1953). The vectors ranked as NP and RP were assigned half-weight. After the first iteration, sample directions falling outside two standard deviations from the initial mean were omitted. The mean characteristic direction for each outcrop is generally the combined results of normal- and reverse-polarity sets, but a vector-sum method that incorporates the mean intensities of the normal- and reversed-polarity vectors was also employed for comparison. In some cases, especially when the relative numbers of normal- and reversed-polarity samples differ significantly from each other and there may be a small residual overprint on all samples, the sum of the mean normal-polarity vector and the reversed-polarity vector can yield a less distorted mean characteristic direction.

The majority of these sections contain an insufficient number of high-quality normal- and reversed-polarity samples to perform statistically significant tests of antipodal directions, therefore, all high-quality data from eight sections were merged to a single composite. Analysis of paleomagnetic directions within that composite also allowed identification of statistical outliers that distorted the results of the small sample suites from the individual sections. The dual polarities of this composite set passed significance tests (Table 2A).

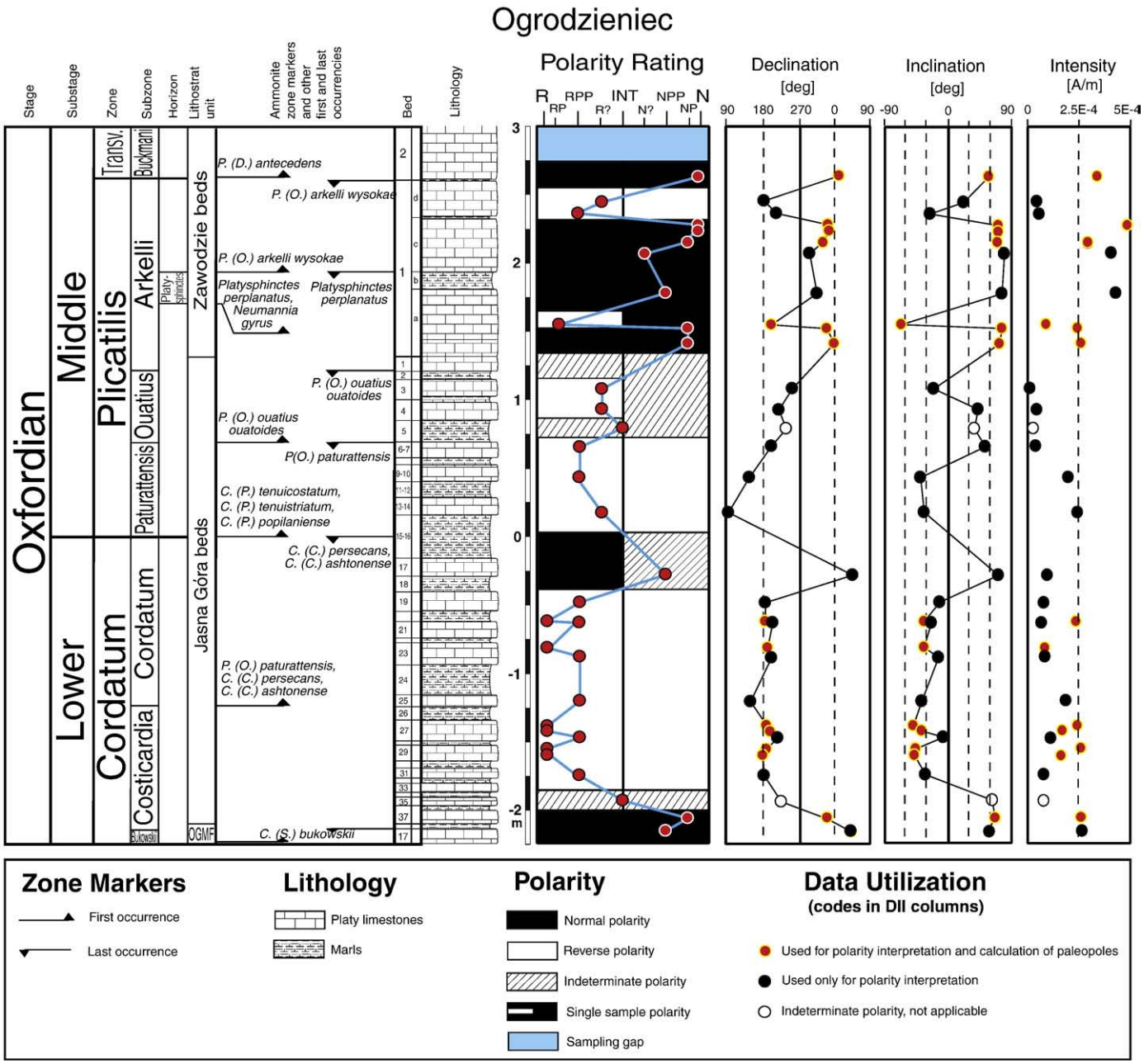


Fig. 5. Ogrodzieniec magnetostratigraphy with placement of main ammonite levels and interpretation of ammonite zonation. Polarity rating: N/R — good normal- or reversed-polarity behavior, INT — indeterminate or intermediate. Intensity column represents the intensity of the computed vector for the characteristic direction of each sample. Abbreviations in ammonite columns: "P." = *Perisphinctes*, "(D.)" = *Dichotomosphinctes*, "(O.)" = *Otosphinctes*, "C." = *Cardioceras*; "(P.)" = *Plasmatoceras*, "(S.)" = *Scarburgiceras*.

5.2. Mean virtual geomagnetic pole of the Lower–Middle Oxfordian

The mean virtual geomagnetic pole of the composite from these Oxfordian sections of Poland is at 78.5°N, 184.9°E ($\delta p = 2.6°$, $\delta m = 3.5°$) (Table 2A). This paleopole is within 4° to a paleopole derived from several Upper Oxfordian–Lower Kimmeridgian sections from the Polish Jura Chain and the mean paleopole for a composite Kimmeridgian–Oxfordian boundary interval for the European craton (Przybylski et al., in press).

This paleopole appears to be rotated counterclockwise about 11°–7° relative to mean paleopoles for Late Callovian or the Late Kimmeridgian intervals of an apparent polar wander paths (APWP) for Europe as estimated by Besse and Courtillot (2002). However, this portion of their European APWP was computed from the least number of sites relative to any other time span in the 200 myr span of their compilation. Nevertheless, the mean paleolatitude (39.3°N) that we derive for our Oxfordian sections (geographical center at 50.4°N, 19.6°E) is similar to projected paleolatitudes projected from the Besse–Courtillot European APWP (Table 2B). The coincidence of this Polish Oxfordian paleopole with an independent mean paleopole for the Kimmeridgian–Oxfordian boundary interval (Przybylski et al., in press, which did not incorporate any of the sections in this paper) suggests that these placements are a more constrained estimate of the European APWP for this brief time interval.

6. Magnetostratigraphic results

6.1. Sub-Mediterranean polarity pattern of the Lower–Middle Oxfordian

The magnetostratigraphies of the Polish sections (Figs. 5 and 6, Suppl. Figs. 5–9) were combined with the revised biostratigraphic ages and reinterpreted polarity ratings for the Podłęże1-3 (PD), Rudno (RD) and Zalas2 sections. The NE Iberian Range section composite from Spain (Juárez et al. (1995) is included, but with the revised biostratigraphic boundaries and interpretation of condensation levels.

Within each ammonite subzone, the polarity patterns were generally consistent among sections that had both a precise biostratigraphy and an abundance of high-quality-rated results. This subset was

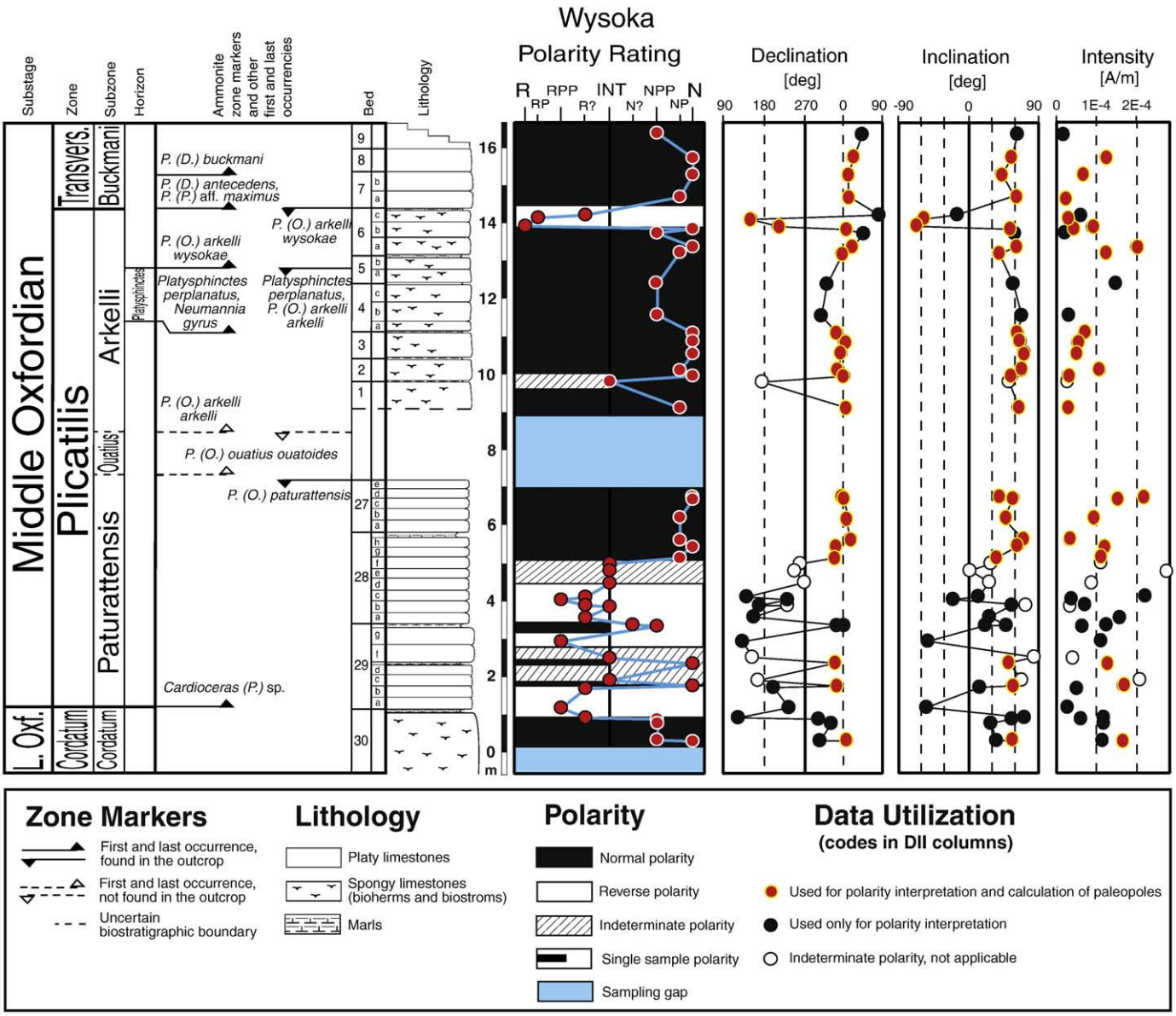


Fig. 6. Wysoka magnetostratigraphy with placement of main ammonite levels and interpretation of ammonite zonation. Polarity rating: N/R — good normal- or reversed-polarity behavior, INT — indeterminate or intermediate. Intensity column represents the intensity of the computed vector for the characteristic direction of each sample. Abbreviations in ammonite columns: "Transvers." = *Gregoryceras transversarium*; "P." = *Perisphinctes*, "(D.)" = *Dichotomosphinctes*, "(O.)" = *Otosphinctes*, "(P.)" = *Plasmatoceras*.

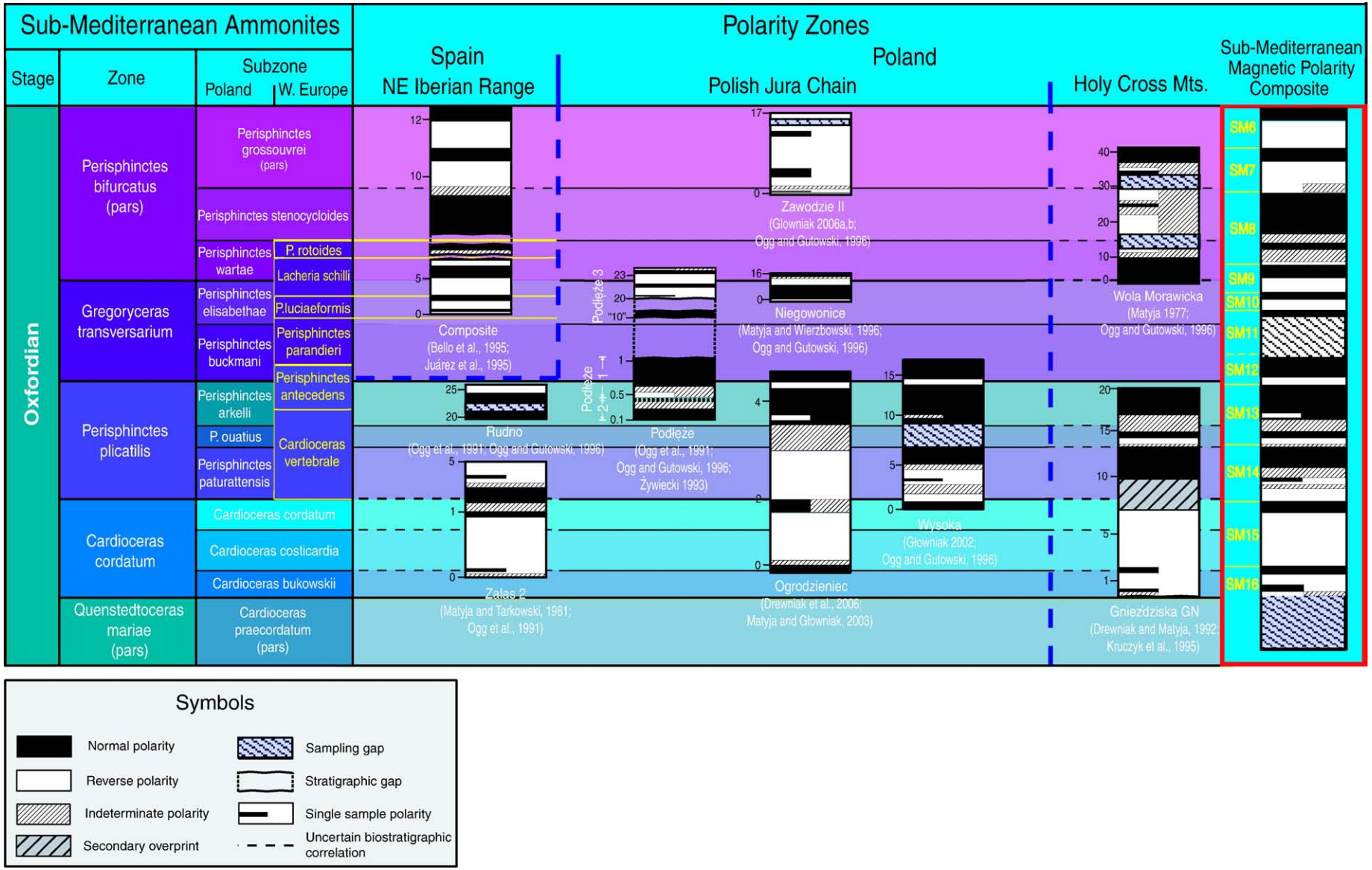
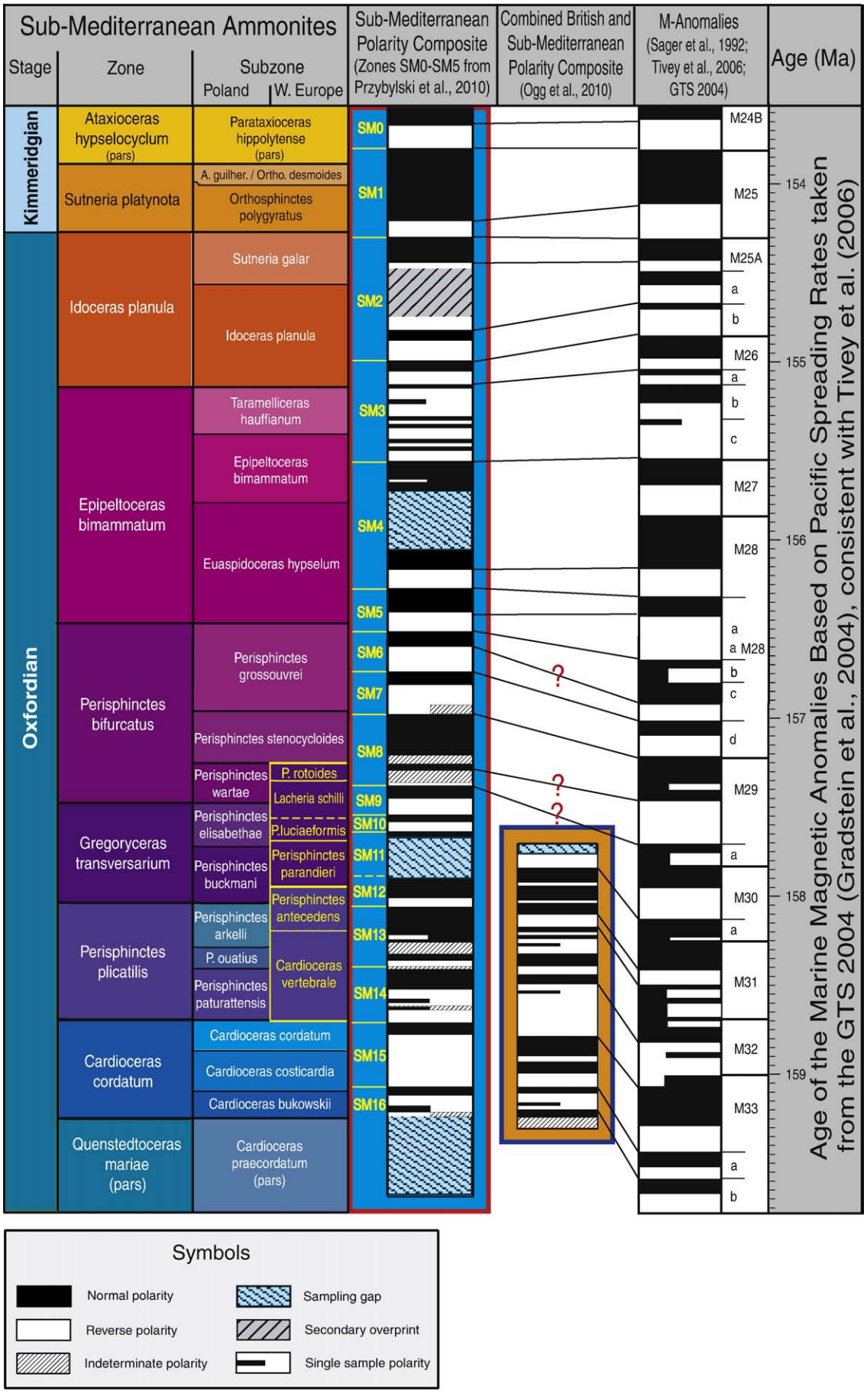


Fig. 7. Sub-Mediterranean magnetic polarity composite of the Lower-lower–Upper Oxfordian derived from Polish and Spanish sections. Ammonite zones are shown here and in Fig. 8 as biozones (nomenclature by Genus–species), rather than chronozones (nomenclature by capitalized Species, as used by many Jurassic ammonite specialists) in order to illustrate possible temporal extent of sedimentary discontinuities among sections. The limits of the *Gregoryceras transversarium* ammonite zone have different definitions in Poland and in Spain (Aguilon); but the inter-regional correlations among the local subzones of the *G. transversarium* and *Perisphinctes plicatilis* ammonite zones are well-defined by common ammonite taxa and datums. Relative durations of biozones and subzones are scaled after cycle-stratigraphy compilations by Boulila et al. (2008. in press). Polish Jura Chain sections of Zawodzie (ZD), Niegowonice (NE), Wysoka (WY) and Ogrodzieniec (OG), and the outcrops from the Holy-Cross Mountains: Wola Morawicka (WM) and Gnie_dziska (GN) are displayed separately (Figs. 5, 6; Suppl. Figs. 5–8). The Sub-Mediterranean polarity composite is constructed from features common to sections having both well-constrained biostratigraphy and higher-quality magnetostratigraphy.


Sub-Mediterranean Ammonites
Stage
Zone
Subzone
Poland
W. Europe
Sub-Mediterranean Polarity Composite (Zones SM0-SM5 from Przybylski et al., 2010)
Combined British and Sub-Mediterranean Polarity Composite (Ogg et al., 2010)
M-Anomalies (Sager et al., 1992; Tivey et al., 2006; GTS 2004)
Age (Ma)
Kimmeridgian
Oxfordian
Ataxioceras hypselocyclum (pars)
Parataxioceras hippolytense (pars)
A. guilher. / Ortho. desmoides
Sutneria platynota
Orthosphinctes polygyratus
Idoceras planula
Sutneria galar
Idoceras planula
Epipeltoceras bimammatum
Taramelliceras hauffianum
Epipeltoceras bimammatum
Euaspidoceras hypselum
Perisphinctes bifurcatus
Perisphinctes grossouvrei
Perisphinctes stenocycloides
Perisphinctes wartae
P. rotoides
Lacheria schilli
Gregoryceras transversarium
Perisphinctes elisabethae
P.luciaeformis
Perisphinctes buckmani
Perisphinctes parandieri
Perisphinctes plicatilis
Perisphinctes antecedens
Perisphinctes arkelli
P. ouatius
Perisphinctes paturattensis
Cardioceras vertebrale
Cardioceras cordatum
Cardioceras cordatum
Cardioceras costicardia
Cardioceras bukowskii
Quenstedtoceras mariae (pars)
Cardioceras praecordatum (pars)
SM0
SM1
SM2
SM3
SM4
SM5
SM6
SM7
SM8
SM9
SM10
SM11
SM12
SM13
SM14
SM15
SM16
M24B
M25
M25A
a
b
M26
a
b
c
M27
M28
a
a M28
b
c
d
M29
a
M30
a
M31
M32
M33
a
b
154
155
156
157
158
159
Age of the Marine Magnetic Anomalies Based on Pacific Spreading Rates taken from the GTS 2004 (Gradstein et al., 2004), consistent with Tivey et al. (2006)
Symbols
Normal polarity
Reverse polarity
Indeterminate polarity
Sampling gap
Secondary overprint
Single sample polarity

used to construct a composite polarity scale that spans nearly the complete Cordatum through Bifurcatus zones (Fig. 7). The lower portion overlaps a polarity pattern for Boreal ammonite zones (Ogg et al., in press), thereby enabling a partial test of consistency of the main features. The main polarity zones are provisionally labeled in a downward continuation of the Sub-Mediterranean ("SM") nomenclature for the overlying Kimmeridgian–Oxfordian boundary interval of Przybylski et al. (in press).

The Mariae Zone of the lowermost Oxfordian is a major sedimentary hiatus in Poland. The Cordatum and lower Plicatilis zones are dominated by reversed polarity with thin normal-polarity intervals recorded in Ogrodzieniec, Wysoka, Gnieździska and Zalas2 (SM16 through lower SM14; Figs. 5–7). This dominance of reversed polarity is consistent with the British results (Fig. 8). However, the relatively condensed Polish sections (e.g. Cordatum Zone spans only 2 m in Ogrodzieniec) may significantly underestimate the number of reversals compared to the British composite (e.g., Cordatum Zone spans nearly 25 m in the Isle of Skye and Yorkshire). The boundary between Cordatum and Plicatilis zones coincides with a narrow normal-polarity band (SM15n).

The Arkelli Subzone of the upper Plicatilis Zone and the lowermost Buckmani Subzone of the Transversarium Zone are dominated by normal polarity with relatively thin reversed-polarity bands in Rudno, Ogrodzieniec, Wysoka, Podłęże-1 and Podłęże-2 (upper SM14 through SM12; Figs. 5–7). Normal polarity is also present in Gnieździska within the upper part of the sampled interval up to the upper Arkelli Subzone. The upper Buckmani Subzone (lower Transversarium Zone) and the base of Elisabethae is a sampling gap (provisionally labeled SM11). The Elisabethae Subzone of the upper Transversarium Zone is characterized by a reversed polarity in its middle part followed by normal polarity in the upper part, as recorded in the Niegowonice section (SM10).

The Elisabethae–Wartae subzonal transition displays dominance by reversed polarity at Niegowonice, Podłęże-3 and at the equivalent stratigraphic interval of the lower Schilli Subzone at NE Iberian Range section composite (SM9). Both the base of the Rotoides Subzone (lowermost SM8) and its top (uppermost SM8) may be discontinuities (Meléndez and Fontana, 1991; Bello et al., 1995, Głowniak, 2006a); therefore, these intervals are indeterminate. The Stenocycloides Zone in the NE Iberian Range composite is a thick normal-polarity zone (SM8). The SM8–SM7 transition at the top of the Stenocycloides Zone in the NE Iberian Range is not well documented due to the equivocal biostratigraphic interpretations of Juárez et al. (1994) versus Juárez et al. (1995), but can be assigned as just below the top of the Stenocycloides Zone at ZawodzieII. The overlying Grossouvrei Subzone is dominated by reversed polarity with thin normal-polarity intervals (SM7 and SM6) recorded by the ZawodzieII and the NE Iberian Range composite.

6.2. Correlation to the marine magnetic anomaly pattern

Magnetostratigraphy studies have progressively correlated the polarity patterns within the Tithonian through uppermost Oxfordian ammonite zones to the more distinctive pattern of Kimmeridgian–Tithonian chrons M25–M19 (e.g., Ogg et al., 1984; Gradstein et al., 2004; Speranza et al., 2005). The diachronous historical uses of the Oxfordian–Kimmeridgian boundary interval in the Boreal, Subboreal and Sub-Mediterranean faunal realms have been correlated to chrons M25 through M27 (Przybylski et al., in press).

The Sub-Mediterranean polarity composite for Lower and Middle Oxfordian must be coeval with a portion of the pre-M26 marine magnetic pattern as interpreted from aero-magnetics and deep-tow magnetic surveys in the oldest part of the Pacific Ocean (Sager et al., 1998; Tivey et al., 2006; Tominaga et al., 2008). A difficulty in deep-tow interpretations is to differentiate which of these geomagnetic signals are from fluctuations of magnetic field intensity and which are from actual field reversals. To partially indicate the range of potential interpretations, the pre-M26 portion of the Pacific M-sequence from deep-tow magnetometer surveys has two models of geomagnetic field behavior — one that assumes the variability observed at 5.5 km depth are small-scale anomalies caused by polarity reversals, and the other is derived from a mid-depth projection of the main features of the magnetic field observations that may underestimate the number of polarity chrons. The "real" pre-M26 geomagnetic polarity pattern may be somewhere between these two models. Geomagnetic time scales were interpolated for both models (Tivey et al., 2006; Tominaga et al., 2008) by assuming a constant-spreading model for the Japanese spreading center between radiometric-dated ODP Site 801C (late Bathonian basalt, marine magnetic anomaly M42, Koppers et al., 2003; Tominaga et al., 2008) and ODP Site 762 (basalts of anomaly M26, within Oxfordian–Kimmeridgian boundary interval, Ludden, 1992).

Expanded strata spanning each of the Polish ammonite subzones or their equivalent subzone in the Western European zonal scheme have been analyzed for cycle-stratigraphy in multiple sections in southeastern France (Boulila, 2008; Boulila et al., 2008, in press). The main exception is the Rotoides Subzone, in which condensation and possible hiatuses have distorted the magnetostratigraphy and probably caused an underestimate of the cycle-counts in France and corresponding estimated duration (Boulila, 2008, in press). This astronomical tuning of the duration of each ammonite subzone enables us to partially adjust for variable sedimentation rates and to merge sections from adjacent ammonite zones onto a common "linear-interpolated time scale" (Figs. 7 and 8) for comparison to the marine magnetic anomaly pattern created at Pacific spreading centers. However, we did not attempt to compensate for possible finer-scale sedimentation changes within the individual ammonite subzones, therefore there will be distortions to the fine-scale magnetostratigraphic pattern relative to the models for coeval marine magnetic anomalies.

The cycle-scaled Sub-Mediterranean polarity pattern is generally consistent with the frequency of the pre-M26 polarity chrons as interpreted from the deep-tow surveys. The exact match depends on which method of interpreting deep-tow marine anomalies is used. Therefore, for our proposed correlation suite, we followed three rules: (a) The main polarity blocks that are common to both the deep-tow and the mid-tow interpretations should have a corresponding dominance of that polarity in the magnetostratigraphy (e.g. normal-polarity block of M29n). (b) The correlation lines between magnetostratigraphy among sets of cycle-scaled ammonite subzones and the constant-spreading-rate M-sequence should be generally parallel; because significant divergent or converging correlation lines would require either jumps in Pacific spreading rates or significant errors in the cycle-stratigraphy. (c) Condensed deposits in Poland (SM16–SM14) and hiatus-bound intervals (e.g., Rotoides Subzone) are considered to record only a portion of the geomagnetic polarity history. This condensation is particularly important for the upper Cordatum Zone

Fig. 8. Tentative correlation of the Sub-Mediterranean Oxfordian magnetic polarity composite with a Lower Oxfordian pattern from Boreal sections in Britain (Ogg et al., in press) and the M-sequence of marine magnetic anomalies. Abbreviations in ammonite columns: "A. guilher./Ortho. desmoides" = *Ataxioceras guilherandense*/*Orthosphinctes desmoides*. Relative durations of biozones and subzones are scaled after cycle-stratigraphy compilations by Boulila et al. (2008. in press). The traditional Sub-Mediterranean placement of the Oxfordian–Kimmeridgian boundary is at the base of *S. platynota* ammonite zone (approximately base of Chron M25r); but the Boreal and Subboreal placement and candidate GSSP for the Oxfordian–Kimmeridgian boundary (coincident bases of *Amoeboceras bauhini* and *Pictonia baylei* ammonite zones) correlates to the middle of the Sub-Mediterreanean *E. bimammatum* Zone (approximate base of M26r) (Przybylski et al., in press). The right side of the M-Sequence column ("M-Anomalies" = Marine magnetic anomalies) is an interpretation of the deep-tow magnetometer observations at 5.5 km depth; the left side is a mid-depth projection that emphasizes the main features (see text). Projected ages of marine magnetic anomalies based on Pacific spreading rates used in GTS 2004 (Geologic Time Scale 2004; Gradstein et al., 2004) are consistent with ages suggested by Tivey et al (2006).

and the Plicatilis–Transversarium transition interval (equivalent to the Antecedens and Parandieri subzones of Britain) therefore a magnetostratigraphy from the relatively expanded British strata (Ogg et al., in press) was given precedence in this part of the correlation. Given these constraints, a possible suite of correlations are suggested (Fig. 8).

The correlation of reversed-polarity-dominated SM3 to chrons M26br–M26c is the base of the the Oxfordian–Kimmeridgian boundary interval compiled by Przybylski et al. (in press). Following the rules of parallel correlation between the linear-age-interpolated patterns, then the chrons M27–M28n interval matches the two normal-polarity zones of SM4 separated by a non-recovered interval, and chron M28a matches polarity zone SM5. The SM6–SM7 correlation to either model for anomalies M28b through M28d is not perfect; but is constrained by the required match of the major normal-polarity zone SM8 to the relatively long normal-polarity chron M29n. A correlation of SM9 with chron M29a is proposed with the implicit assumption that the condensed Rotoides Subzone was underestimated by the cycle-stratigraphy of France, and its hiatus-delimited strata yielded an inadequate record of the reversed-polarity chron M29r.

Below the upper Transversarium Zone, the current Sub-Mediterranean polarity pattern derived from condensed sections is inadequate to propose a full set of correlations. The normal-polarity dominance of SM12–SM14n is consistent with the mid-tow level model for chrons M30a–M32n. An independent suite from Britain provides a record of a significant normal-polarity zone of upper Antecedens and lower Parandieri, partly filling a gap in the Sub-Mediterranean suite and which fits with the major normal-polarity chrons M30a–M31n.

There is an apparent mismatch of the polarity dominance between the British composite and Sub-Mediterranean polarity zones SM13–SM14n; however the Polish composite may include artifacts of an incomplete removal of a normal-polarity overprint from the Gnieździska2 and in part of the Gnieździska sections (see discussion in section 4). The British composite is consistent with the deep-tow interpretation of marine magnetic anomalies for chrons M31r–M32n. The reverse-polarity-dominated lower portion of the Plicatilis Zone (SM14r) projects to chron M32r using the “parallel-line rule”.

The reversed-polarity-dominated SM15 is mainly based on condensed strata in Ogrodzieniec, whereas the British composite from relatively expanded strata is consistent with chron M33n. The “parallel-line rule” projects lower polarity zone SM15r and SM16 to correlate with chrons M33a and M33b.

Therefore, in general, the Middle Oxfordian composite (Transversarium through Bifurcatus) is correlated with the interpreted marine magnetic anomalies M28b through M31; and the upper-Lower Oxfordian (augmented with results from Britain) is consistent with M31r through M33b.

Our Oxfordian magnetostratigraphy supports the interpretation that these low-amplitude pre-M26 marine magnetic anomalies are recording geomagnetic field polarity chrons (Sager et al., 1998; Tivey et al., 2006; Tominaga et al., 2008). Portions of the outcrop-based magnetic polarity scale of this study enable evaluation of the two methods of interpreting Pacific anomalies for different intervals.

7. Conclusions

Magnetostratigraphic results from several ammonite-rich sections in southern Poland are merged with those from the Iberian Range in Spain yield a nearly continuous polarity pattern for the middle two-thirds of the Oxfordian (Sub-Mediterranean ammonite subzones of Cordatum through Bifurcatus Zones). Significance tests and rock-magnetic experiments of selected lithologies necessitated rejection of some paleomagnetic data that had a probable overprint of a secondary magnetization. A similar intensive study of all lithologies for each polarity zone at each outcrop would be desirable to increase the reliability. The mean paleopole of the suite of Oxfordian sections of Poland at 78.5°N, 184.9°E ($\delta p = 2.6°$, $\delta m = 3.5°$) is within 4° of the Oxfordian–Kimmeridgian pole from the European craton (Przybylski et al., in press).

The composite pattern of 11 main polarity intervals (SM6 through SM16) is scaled to the durations of each ammonite subzone as calibrated by cyclostratigraphy (Boulila et al., 2008, in press) and the lower Oxfordian interval is partially augmented by coeval sections in Britain. This cycle-constrained magnetic polarity scale confirms that marine magnetic anomalies M28 through M33 represent geomagnetic reversals. With additional magnetostratigraphy studies of more expanded sections that have both a similar precise biostratigraphy and are amenable to cycle-stratigraphic scaling, then we should be able to develop a reliable Oxfordian polarity time scale to better model and date the indistinct pre-M26 marine magnetic anomalies of the Pacific Ocean.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.epsl.2009.11.030.

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